

Kanad Motiwale

Boston, MA (Open to Relocation) | 617-708-7658 | motiwale.k@northeastern.edu | LinkedIn | GitHub | Portfolio

Education

Northeastern University

Master of Science in Computer Science (GPA: 3.93/4.0)

Sep 2025 – Dec 2027

Boston, MA

Ramrao Adik Institute of Technology, University of Mumbai

Bachelor of Engineering in Electronics & Telecommunication (GPA: 3.5/4.0)

Aug 2019 – May 2023

Mumbai, India

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, Java, C++, SQL, R, HTML5, CSS3, Bash

Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, pandas, NumPy, Drizzle ORM, Selenium, OpenCV

AI / ML: LLMs, RAG, Agentic AI, OpenAI API, FAISS, Pydantic AI, NLP, Vector Databases, Prompt Engineering

Data Engineering: ETL/ELT, Dimensional Modeling, Data Warehousing, PostgreSQL, MySQL, SQLite, Query Optimization

Cloud & DevOps: AWS (Lambda, EC2, S3), Docker, GitHub Actions, Azure DevOps, CI/CD, Git

Practices: REST APIs, Microservices, Distributed Systems, MVC, TDD, Unit/Integration Testing, Agile, System Design

Experience

Northeastern University – Khoury College of Computer Sciences

Jun 2026 – Present

Graduate Teaching Assistant

Boston, MA

- Mentor 60+ students across DS 2500 (Intermediate Programming with Data) and DS 3500 (Advanced Programming with Data) building production-grade Python systems and REST APIs with FastAPI, SQLAlchemy, pytest, and Redis.
- Run 50+ code-review meetings on FastAPI services and service-oriented architectures, auditing Git history and testing.
- Lead office hours and debugging on ETL, SQLAlchemy (SQLite/PostgreSQL), Redis, and pytest; grade projects on architecture.

Northeastern University – D'Amore-McKim School of Business

May 2026 – Jul 2026

Graduate Research Assistant

Boston, MA

- Architected a multi-agent AI platform integrating GPT-5.5, DeepSeek, and Llama-3.3-70B through a unified orchestration layer, enabling collaborative and competitive reasoning workflows from a single user request.
- Engineered a resilient LLM inference infrastructure by implementing request timeouts, automatic retries, and cross-provider failover strategies, improving reliability under concurrent workloads and maintaining stable operation across 130+ production deployments.
- Built a behavioral analytics platform using Vercel KV (Redis) to capture session telemetry, edit provenance, and user engagement metrics, delivering an admin dashboard with interactive visualizations and Excel/JSON export capabilities.

LTM

Dec 2023 – Aug 2025

Software Engineer

Mumbai, India

- Built and enhanced full-stack functionality for Marsh's Policy & Placement Management platform using MERN Stack and Unqork, supporting policy administration workflows and complex layer/coverage data processing.
- Designed automated test suites and validation frameworks for JSON-driven insurance workflows, reducing manual testing effort by 60% and accelerating release cycles across multiple product modules.
- Diagnosed and resolved 20+ critical production defects through root cause analysis, debugging, and performance testing, decreasing issue resolution time by 45% and improving application reliability.

FastFindFirm

May 2023 – Oct 2023

Software Developer Intern

Mumbai, India

- Built web and mobile applications for an ed-tech platform serving 5,000+ users across online learning and internship programs.
- Developed full-stack applications using React, React Native, Node.js, and MongoDB, supporting 1,500+ students across cohorts.
- Built platform features, Dockerized services, and CI/CD pipelines for 10+ internship programs and 300–400 student cohorts.

LTM

Feb 2023 – Apr 2023

Software Engineer Intern

Mumbai, India

- Developed backend modules and REST APIs in Python and Java using OOP principles across a production-scale codebase.
- Authored 40+ unit and integration tests via TDD, reducing API error rates by 18%; performed root cause analysis on defects.
- Collaborated in Agile/Scrum ceremonies on code reviews and feature delivery; received full-time return offer upon completion.

Projects

Lexorion | Python, FastAPI, FAISS, Ollama, Pydantic AI, React, SQLite | [GitHub](#)

Apr 2026

- Built a fully local RAG pipeline using FAISS, Ollama (llama3.2), and nomic-embed-text embeddings with zero cloud costs.
- Designed an ETL pipeline in FastAPI to parse PDFs, chunk text with tiktoken, and index embeddings into FAISS with SQLite.
- Implemented a Pydantic AI multi-agent re-ranking layer; built a React frontend with chat, upload, and search panels.

SponsorFlow | Next.js, TypeScript, PostgreSQL, Drizzle ORM, NextAuth.js, Tailwind CSS | [GitHub](#)

Mar 2026

- Architected a full-stack sponsorship marketplace connecting event organizers and businesses using Next.js 16 and PostgreSQL.
- Implemented role-based authentication via NextAuth.js v5 and Drizzle ORM across 8 relational tables.
- Delivered tiered package management and proposal tracking with Tailwind CSS, Zod validation, and Zustand.

Calendar Application | Java, OOP, MVC Architecture, Data Structures, Algorithm Design, JUnit, TDD | [GitHub](#)

Dec 2025

- Built a multi-mode calendar app in Java supporting multiple time zones, recurring events, and GUI, CLI, headless modes.
- Applied OOP design patterns and MVC architecture to implement scheduling algorithms for time zone and recurrence handling.
- Achieved 90%+ code coverage with 60+ JUnit tests via TDD and optimized data structures across all execution modes.